

José R. Sousa

Charlotte, NC 28217

LinkedIn: <https://www.linkedin.com/in/jrsousa2/>

Phone: 904-290-2890

E-mail: jrsousa2@gmail.com

GitHub: <https://jrsousa2.github.io/>

SUMMARY OF QUALIFICATIONS

- Large experience in **database / data warehouse** and **SQL** (query language).
- Great with **Python** programming in **VS Code**. Created executable from code, developed **REST API's**, created **GUI's**, , loaded **Excel** and **XML** files into **pandas** dataframes, read and wrote to Excel files with **openpyxl**, created formatted **PowerPoint** files automatically, created **virtual environments**, automated **batch jobs**, etc. **All can be viewed on my GitHub**.
- Extracted data from **Hadoop** with Dremio / Hive and created VDS's.
- Fitted **logistic regression** models using both **SAS** and **PySpark ML**.
- Created **GLM** to predict bare land value of properties in Orange county, CA.
- Independently created **logistic regression** model to score risk of auto policyholders. The model was deployed.
- Worked on **insurance pricing models** in partnership with consultancies.
- Created **workflows** in **n8n** for Gen AI.
- **Expert at SAS** (**Enterprise Guide** and **Viya**) -- created **stored processes** and **ETL's**, wrote advanced code to produce tables and reports, and truly mastered **SAS macros**. Created dashboard with **GRAPH** and reports with **Proc Report**.
- Strong with **Machine Learning** models and **data analytics** and familiar with **actuarial analyses** and **actuarial reports**.
- Expert at **MS Excel**, created advanced spreadsheets with **VBA macros**, **automated charts and graphs**.
- Most of my experience is in the **insurance** industry (15+ years). Created a wide variety of **loss triangles**, estimated **ultimate claims** from LDF's, calculated ceded losses from **reinsurance** contracts, calculated earned premiums, created reports on **loss ratio** and **in-force policies**, built **predictive models**, etc.
- Exposure to **accounting principles** – using the **GL** Oracle database, created **YTD summary reports** with array formulas in **Excel** to support quarterly filings (10-Q and 10-K) and performed account-level reconciliations with it.
- Graduated in **Math / Statistics** from the **best university in Brazil** (**U of São Paulo**).
- Very analytical, self-taught, can **work independently**, created new formulas in math (e.g. <https://arxiv.org/html/1810.07877v7>).
- **U.S. citizen** (had a **green card** through **NatGen** previously).

SOFTWARE

- **Databases:** **Oracle**, **Teradata**, **SQL Server / SSMS** and their **IDE's**.
- **Big Data:** **Snowflake**, **Databricks**, **PySpark**, **Hadoop**, **Dremio**, **Hive**.
- **Programming:** **Python**, **VBA**, **VBScript**, **Unix/Windows batch scripting**.
- **Cloud:** Working knowledge of **AWS**, **Azure** and **GCP**.
- **SAS:**
 - Platforms: **Unix**, **Linux**, **Windows Server** and **Viya** (Cloud).
 - Modules: **BASE**, **STAT**, **GRAPH**, **ACCESS**, **CONNECT**.
 - Expert knowledge of most used SAS Procs, including **Proc Report**. Advanced **macro language** and **stored processes**.
- **BI / Predictive Modeling:** **SAS Enterprise Guide**, **Power BI**.
- **Code Editors:** **VS Code**, **Jupyter Notebook**.
- **Other:** **Excel**, **PowerPoint**, **SharePoint**, **Git**, **RegEx**, **Jira**, **XML**, **JSON**.

PROFESSIONAL EXPERIENCE

Wells Fargo (banking)

Senior Data Analyst / Data Scientist (contract / hybrid)

10/2022 – 12/2025

▪ **Second project: 2025**

- Worked on **AML / Financial Crimes** remediations (CIP / BO / CDD / **KYC**).
- Built AML remediation populations (Trust, NRM4 and transportation assets).
- Created population identification workbooks and waterfall charts and presented them to senior directors for approval.
- Wrote complex **optimized SQL** queries to extract data from **Hadoop** using **Dremio** and **Hive** and from various databases (**Teradata**, **Oracle**, SQL Server, DB2, etc.). Was a go-to person for **challenging** technical questions.
- Developed **SAS** and **SQL** code to create ad-hoc, time-sensitive data extracts and **reports**.
- Created code test scripts and worked with code testers to validate the code.
- **Recognized with a full-time offer at project completion.**

▪ **First project: 2022 through 2025**

- Worked on **customer remediations**, responsible for the creation of the final artifacts from which refunds and letters are issued to customers as required by **regulators**, following strict **Confluence** procedures.
- Managed **SAS** code used to create final remediation artifacts, made changes to the code to add **enhancements**, **optimize performance** and minimize the number of **defects** and findings in **peer reviews**. As an example of enhancement, when a new analyst takes over a remediation, all the **artifacts** and **SAS** tables created by the previous analyst are **automatically** saved to a new subfolder as **backup**.
- Created **ETL**'s to produce new artifacts by mapping fields from the execution data requirements (EDR) to data tables. Contributed to design **artifact layouts**.

Truist, Charlotte, NC (banking)

Data Analytics SQL/Python (contract / remote)

09/2021 – 03/2022

- Created rules based **decision tool** for the compliance and data analysts to choose their optimal analytics application as a web form created in **Python**.
- Created and maintained current report on user accesses for the various **proprietary applications** and data repositories (**AD**, **ACL**), for **compliance**.
- Ran reconciliation between **ServiceNow/RITM** and data pulled from the **mainframe** to assess correctness of an automatic ticket creation process for abends (abnormal job termination), and presented findings with **PowerPoint**.
- Ran **SQL** queries in **Snowflake** to extract data from the **data lake/data warehouse**.
- Created reports using the **SAS grid** and produced **documentation**.

Florida Blue, Jacksonville, FL (healthcare)

SAS/Python Developer (contract / remote)

04/2020 – 08/2021

- Developed **ETL** code to extract data from **Oracle** and **DB2** and others, using **SAS 9.4** on **AIX** (Unix) and **Enterprise Guide**.
- Extracted data from multiple sources to create agent **fraud detection** analytics for the **SIU** dept with various metrics. Created **macro to automate** the formatting of the report for each metric using **Proc Report**.
- Automated and scheduled various **SAS ETL jobs** in **Unix**, using a **SAS macro**.
- Used **HTML** syntax in **SAS** to compose rich text e-mail alerts.
- Combined Windows **batch**, **WinScp** and **VB** scripts to schedule jobs and automate file transfers and moving data between platforms using SFTP.
- Created a neat **SAS stored process** for various canned reports.
- Created **visualizations** and **reports** using **Power BI** and **Python**.
- Gave **training** to team members on how to design **SAS stored processes**,

covering how to create the GUI and how to handle the variable user's inputs.

National General (Allstate), New York, NY (insurance)
Insurance Data Programmer / Data Manager

04/2015 – 12/2019

- Reported to the holding company's chief actuary.
- Mastered the claims data system of the company (EPIC) and legacy data stored in Oracle and SQL Server databases.
- Planned and created datasets in SAS by extracting and combining data from various databases, from which loss triangles and data analytics were produced.
- Created actuarial loss triangles used in reserving analyses with SAS macros, and reconciled the numbers with the company's general ledger.
- Created quarterly company-wide YTD summary report in Excel using general ledger data extracted with Oracle BI Discoverer to support quarterly filings and the annual report preparation (10K, 10Q), for the SEC and other regulators.
- Participated in the ATLAS project, with a goal to capture claim and policy level data into the GL Oracle database.
- Extracted data from the SAS server to create reports with SAS macros and code. These reports were used internally or provided to consultants, auditors and regulators.
- Using SAS, ran statistical data analyses to track performance of products.
- Created complete documentation about work done on legacy data from acquired companies.

AIG, New York, NY (insurance)
Data Scientist / SAS Programmer (contractor / FTE)

12/2011 – 12/2014

- Planned and extracted data from multiple sources (mainframe, DB2, SQL Server and Oracle) to produce enhanced tables used in all reports and models and processes, using ETL and SAS/Access interface to OLE DB and ODBC.
- Created ETL's to read VB/VSAM files on IBM mainframe and load them into SAS tables on Windows. These files became the data foundation of the Global Actuarial Pricing department.
- Wrote SAS macro to create a wide variety of loss triangles used to derive LDF's and estimate the ultimate claims.
- Worked in the deep dive and profitability analyses of the book of business, by extracting data on losses, premium, policy and insured attributes to generate various reports and metrics and to monitor the performance of the various lines of business and segments.
- Created smart algorithm for linking policies based on insured and policy characteristics in SAS. This policy linking was the basis for the creation of a scoring model called AQI (Account Quality Index).
- Worked with Deloitte consultants and provided them with loss triangles used as inputs into the Commercial Auto pricing tool. Also worked with Deloitte on the Specialty Worker's Comp model, making sure that deliverables such as SAS code, data and results of the predictive model met the requirements.
- Created and scheduled weekly jobs to extract submission data entered by underwriters into the AQI model and update a Qlikview dashboard.

EDUCATION

Masters's in Statistics - University of São Paulo, Brazil

Bachelor's in Statistics - University of São Paulo, Brazil

Bachelor's in Mathematics - University of São Paulo, Brazil